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Assessing the effects of piglet behavior, diet supplementation, and lesion score breeding values of sires on behavior pattern in weaned pigs

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HIGHLIGHTS

- A herbal diet supplement implying a diet enriched with crude fiber, protein and chicory, was associated with less aggressive pig behavior in the post-weaning period.
- Early piglet behavior was related with respective behavior pattern in the post-weaning period.
- Correlations between breeding values for lesion scores (victim perspective) with video behavior pattern reflecting the actor perspective were only weak to moderate.
- Video images can be used to monitor active pig behavior in the post-weaning period, but it is very challenging to include a large number of pigs due to the time consuming complexity.

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ABSTRACT

The aim of the present study was to study effects of feeding in terms of a herbal diet supplement (HD; basal diet plus a supplement with chicory herbs), of early piglet backtest scores and of relative breeding values for skin lesions (RBV-LS) on behavior pattern of weaned pigs during the post-weaning period. In this regard, we implemented a balanced research design, allowing a semi-randomized allocation of pigs in groups with a very similar number of observations for all sub-cells of diet x backtest score x RBV-LS class combinations. With regard to backtest scores, piglets were classified as high-resisting (HR), low-resisting (LR) or intermediate-resisting (IR). RBV-LS were estimated for the sires based on skin lesions from 993 offspring. The sires were categorized into 2 groups with RBV-LS $>$ 100 (favorable genetic value indicating only a few or no lesions) or RBV-LS $\leq$ 100 (indicating a large number or severe lesions). Video images were analyzed one day after weaning and 5 weeks later, and considered 300 min per monitoring date for the 8 different possible behavior traits resting time (REST), body contact (BCON), initiating fights (IFIGHT), fighting (FIGHT), refusing of fights (RFIGHT), ear or tail biting (BITE), explorative behavior (EXPLORE) and remaining activities (RACT). Finally, 104 pigs with complete observations over the total video monitoring time at both recording dates, and with complete observations for the explanatory variables sire RBV-LS, backtest score and feeding group, were considered for ongoing association studies. The 104 pigs were crosses from matings of Piétrain boars with German Landrace or German Edelschwein sows. The 104 pigs were offspring of 9 different boars (sire lines). Effects of feeding, backtest score and RBV-LS class on video behaviors were inferred via mixed model analyses, implying 8 different runs for the 8 video behavior traits. Least squares means (lsmeans) significantly ($P < 0.05$) differed between HD and the control feeding group (CON) for FIGHT and BITE, with longer durations of aggressive behavior for CON pigs. Vice versa, lsmeans for REST indicating calm behavior were larger for the HD group. Pigs classified as HR piglets were more aggressive than the LR and IR contemporaries, with significantly higher lsmeans for FIGHT, BITE and EXPLORE. The pigs allocated to the sire RBV-LS $>$ 100 group had significantly ($P < 0.05$) longer durations for REST and RFIGHT. Vice versa, pigs allocated to the sire RBV-LS $\leq$ 100 groups spent more with aggressive behavior in terms of BITE and FIGHT. Consequently, boar RBV-LS were favorably correlated with REST, IFIGHT and RACT. In

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conclusion, the early piglet backtest scores as well as breeding values from a victim perspective (lesion scores) can be used as indicator traits for selection against aggressiveness. Nevertheless, results from the present study are first indications and breeding value correlations are approximations in this regard, which should be validated in ongoing studies based on larger datasets via real genetic correlation estimates between actor and victim traits, and considering social interactions and interactions between all fixed effects simultaneously.

1. Introduction

Aggressiveness of pigs including biting is part of natural dominance behavior (e.g., Taylor et al., 2010). However, tail biting in pigs is a severe abnormal behavior with multi-factorial origin and wide ranging consequences, addressing economic aspects and animal welfare. Even though tail biting is mainly observed in intensive production systems (e.g., Thodberg et al., 2018), it also appears in outdoor herds (Walker and Bilkei, 2006) and under organic conditions (Hansson et al., 2000), but in lower prevalence. The routinely use of tail docking in piglets, an effective procedure to reduce tail biting in all pig production systems, is banned in the EU according to the EU Council Directive 2008/120/EC from December 2008. Consequently, it is imperative to evaluate alternative methods addressing feeding and breeding strategies.

The impact of feeding includes various aspects, e.g., food quality or type of food presentation. In this regard, the natural need of rooting and chewing was identified as an important factor to reduce tail biting (Sonoda et al., 2013), which cannot be compensated when feeding a liquid diet (Temple et al., 2012). Aikins-Wilson et al. (2022) reported a lower prevalence for lesion scores in piglets fed an herbal diet with higher crude fiber and crude protein contents compared to the control group fed a basal diet. In contrast, Hunter et al. (2001) found higher levels of tail injuries in pigs fed pelleted when compared to pigs receiving meal or liquid feed. Fraser et al. (1991) and Fraser (1987) found that food diets with inadequate protein or mineral content attract pigs to blood. Diets enriched with herbs implied less aggressiveness and fewer tail and skin lesions (Jensen et al., 1993; Kallabis and Kaufmann, 2012). In addition, endotoxins generated in case of high-density energy feeding rations caused ear and tail necrosis with acute itchiness (Jaeger, 2013). Due to this unpleasant side effect, affected pigs may be attracted to or even tolerate biting attempts by pen mates.

With regard to breed effects, Westin (2003) reported differences in the propensity to bite (Landrace > Yorkshire > Hampshire) as well as in the propensity to be bitten (Yorkshire > Landrace > Hampshire). Accordingly, Sinisalo et al. (2012) identified Yorkshire pigs as victims more often than Landrace pigs. In other studies (Lund and Simonsen 2000; Guy et al., 2002), the breed effect on tail biting was non-significant. Aikins-Wilson et al. (2021) outlined the possibilities to use skin lesion scores to select against aggressive pigs. Oppositely, based on comprehensive video analyses for pig behavior combined with data for skin lesions, Liu et al. (2022) highlighted the effects of standoff and being bullied on lesions rather than the possible cause of unilateral active aggressive behavior. Regarding feeding x genotype interactions, Aikins-Wilson et al. (2022) identified significant sire x diet interactions (herbal diet vs. basal diet) on lesion scores, and they suggested specific boars for different feeding environments. However, from an actor or biting perspective, there is a lack of knowledge addressing interactions between feeding with breed, sire line or genotype effects.

Effects of the pigs' personality and coping style to aversive stressors, evaluated through associations between behavioral tests before weaning with biting behaviors later in life, are unclear (Prunier et al., 2019). First hints in this regard were outlined in some previous studies (Hessing et al., 1993; D'Eath and Burn, 2002; Melotti et al., 2011). Turner et al. (2008) and Hessing et al. (1993) performed behavior backtests in piglets (according to the protocol by Zebunke et al., 2015). Abnormal piglet behavior was associated with aggressiveness later in lifer, suggesting the use of early backtest scores as indicator traits in pig selection schemes. Prunier et al. (2019) addressed the complexity of this topic including the aspects and the interplay of coping style, personality and biting behavior, and they indicated the challenges of behavior trait recording. Several electronic techniques, such as location detectors (D'Eath et al., 2014), activity measurement devices (Dawkins et al., 2009) and 3D-cameras (Scotland's Rural College (SRUC), 2022), have been used to detect abnormalities in pigtail postures automatically. Changes in tail postures and in behavior pattern can be used as early warning indicators for tail biting outbreaks. Specifically, Zupan et al. (2012) used heart-rate belts to detect differences in heart rates between pigs performing tail biting, pigs that were victims and pigs that were not involved in tail biting at all. Such automatic techniques are efficient to identify biters, but costly. D'Eath et al. (2014) favored the analysis of comprehensive video images or a combination of modern video technique with machine learning algorithms to identify biters, but they also indicated the time consuming aspects in monitoring video images.

Consequently, taking into account all above-mentioned challenges, the aim of the present study was to infer the effects of early piglet behavior, feeding, boar line and breeding values for skin lesions (indicator trait for a bitten pig) on abnormal behavior pattern monitored via video images of pigs during the post-weaning period.

2. Materials and methods

2.1. Research design, feeding and housing conditions

The experiment was conducted at the research farm Oberer Hardthof of the Institute of Animal Breeding and Genetics, University Giessen, Germany, over a period of 12 months from March 2018 to February 2019. The piglets with later observations after weaning for video behavior images used in this experiment were a sub-sample from 993 piglets as considered in the study by Aikins-Wilson et al. (2022) for the estimation of breeding values for lesion scores. The piglets were offspring from matings of Piétrain boars with German Landrace or German Edelschwein sows. The piglets were weaned 28 days after birth. At weaning, 112 piglets were semi-randomly allocated to two different feeding groups either receiving a basal diet (CON, $N = 56$ pigs) or an herbal diet (HD, $N = 56$ pigs). The semi-randomly grouping approach aimed on an equal number of offspring per boar (= sire line) and of an equal distribution of piglet behavior scores in both feeding groups. In this regard, we used the stratify.R function (Reddy and Khan, 2021) from the R-software package. The 112 selected pigs were offspring of 9 different boars (sire lines), implying 13 pigs per sire line. In the post-weaning phase, a sample of 104 pigs could be used for video image analyses.

The feed ration in both groups after weaning is the same as described in the previous skin lesion study (Aikins-Wilson et al., 2022). Hence, in the CON group, pigs received a standard diet based on wheat, soybeans and barley as concentrates plus minerals and vitamins. The HD treatment was the standard diet plus a mixture of 0.2% "Kräuterkraft Aufzucht & Verdauung". The composition of the supplementary feed "Kräuterkraft Aufzucht & Verdauung" in the HD diet consisted of 30% chicory (*Cichorium intybus*) root, carbonic acid algae lime (*Lithothamnion calcareum*), alumroot (*Heuchera americana*), 5% yarrow herb (*Achillea millefolium*), fenugreek seeds (*Trigonella foenum-graecum*), nettle herb (*Urtica dioica*), malt sprouts, sugar beet molasses, yeast and seaweed (*Ascophyllum nodosum*) flour. During the experiment in the post-weaning period, the pigs were kept on a partially slatted floor (0.38 m^2 per animal) and the pig: feeding place ratio was 1.5:1. Each of the eight fattening compartments comprised 14 pigs. The fattening compartments were separated among each other through solid pen walls. The pigs had ad libitum access to dry food and water.

2.2. Animal traits

2.2.1. Behavior test of piglets and lesion scoring after weaning

At birth and at the age of 7 days, a backtest score (BTS) was performed on the 954 piglets using procedures as described by Hessing et al. (1993). During the backtest, the piglets were lifted from their pen on a mat lying on a table. The experimenter turned the piglets on their back by fixing the piglets legs for 60 s with both hands. Based on the number of escape attempts (struggling with at least the hind legs), the piglets were classified into high-resisting (HR = more than 2 escape attempts, indicating aggressiveness), low-resisting (LR = less than 2 escape attempts, indicating calm pigs) or intermediate-resting (IR= exactly 2 escape attempts, indicating intermediate aggressiveness) groups.

Skin lesions were determined by one trained person at day 1 after weaning and 5 weeks later following the protocol by Pluske and Williams (1996). The lesion score (LS) was assigned as follows: LS 1= no lesion or less than 5 slight skin lesions; LS 2= mild lesion with more than 5 mild lesions including hair loss, redness, irritation, scratches or small abrasions; LS 3= severe lesion, bleeding and loss of tissue. The LS phenotypes were used for the estimation of LS breeding values in our previous study by Aikins-Wilson et al. (2022). Hence, the nine Piétrain sires had LS breeding values based on all phenotyped offspring and related animals ($N = 993$) as considered for lesion scoring. The LS breeding values were standardized to obtain relative breeding values for LS (mean= 100; SD= 12). The relative LS breeding values (RBV-LS) of the nine sires ranged from 81 (indicating a large number of lesions) to 133 (indicating only a few or no lesions). According to the threshold for the RBV-LS of 100, the 9 sires were allocated to two groups (5 sires with RBV-LS > 100, and 4 sires with RBV-LS ≤ 100).

2.2.2. Behavior video images of weaned pigs

Behavior of the 112 pigs after weaning was monitored at two dates using a 24-time lapse video camera. Cameras were placed above the pen and connected with a laptop. The first video monitoring date over a period of 5 h was one day after weaning, and the second video monitoring date also considering the period of 5 h was 5 weeks after weaning. The recording time was scheduled between the morning and the evening feeding times from 10:00 am to 3 pm. We also made video observations 2 and 4 days after these two recording dates, and we calculated the accuracy of prediction (r) with repeated records compared with single records (according to Mrode 2005) as follows:

$$r = \sqrt{\frac{1}{t + \frac{(1+t)}{n}}}$$

with t = repeatability and n = the number of repeated measures per pig. For a repeatability of 0.11, the extra gain in prediction accuracy from 2 to 4 measurements in relation to the prediction accuracy from a single measurement was moderate (+ 39%). However, due to the complexness in statistical modelling approaches (see more details in the statistical modelling section), we considered the two measurements, i.e., from the weaning date and 5 weeks later in the ongoing analyses.

104 pigs had complete video images with two repeated observations for all behavior traits. These 104 pigs were considered for the ongoing analyses. The trait categorization of the video images considered the behavior trait definitions according to Statham et al. (2009) as outlined in Table 1. Hence, the 8 analyzed behavior traits included resting time (REST), body contact (BCON), initiating fights (IFIGHT), fighting (FIGHT), refusing of fights (RFIGHT), ear or tail biting (BITE), explorative behavior (EXPLORE) and remaining activities (RACT). Identification of specific behaviors and changes of behavior pattern when analyzing the video images based on protocols by Samarakone and Gonyou (2009). Due to the low frequencies of ear and tail biting, BITE considered both traits simultaneously. The time per behavior trait category and piglet was measured in minutes. A total monitoring time of 5 h implied 300 min for 8 different possible behavior traits per recoding date and pig. The total recording time for the 104 pigs and both recording dates as used for video image analyses comprised 62,400 min. All videos were analyzed by the same trained observer. From 8 pigs considered in the original research experiment, videos were difficult to follow, i.e., due to ambiguous animal identification over the full 5 h recording period. These 8 pigs were excluded from the ongoing analyses. The distribution of pigs used for video image analyses for all possible feeding x backtest score classes is given in Table 2. The applied chi-squared test for frequencies in the different feeding x backtest score sub-cells indicated non-significant differences ($P = 0.15$), supporting the randomized and balanced setup of the research design. With regard to all sub-cells of Table 2, 50% of pigs had a sire LS breeding value > 100, and 50% of pigs had a sire LS breeding values ≤ 100.

Table 1
Description of the recorded behavioral traits based on video images.

Behavioral Trait	Abbreviation	Description
Resting time	REST	Lying, sitting or standing
Body contact	BCON	Light touch to body of opponent with nose without force; no indications of aggressiveness, contact did not result in a further fight; contacts reflect curiosity
Initiating fights	IFIGHT	First attempt to provoke a fight; active aggressive behavior of a candidate resulting in a fight, or resulting in an avoiding distance of another pig.
Fighting	FIGHT	Head to head knocks, head to body knocks
Refusing fights	RFIGHT	Pig receives aggression, but does not react
Ear or tail biting	BITE	Any contact with the tail or ear of the pen mate
Explorative behavior	EXPLORE	Occupation with the toy (metal chain), sniffing with the nose, touching the body of another pig with snout, belly-nosing, genital nosing
Remaining activities	RACT	Feeding, drinking, excretion, running with the head up, and all other general behaviors not listed above

2.3. Statistical analyses

The non-Gaussian behavior traits REST, BCON, IFIGHT, FIGHT, RFIGHT, BITE, EXPLORE and RACT were transformed to achieve normality using Blom's method (see Solomon and Sawilowsky, 2009) as implemented in the function blom.r of the R-software package (R core Team, 2020). Mixed model analyses as implemented in the lme4 function from the lmerTest R-package (Bates et al., 2015) was applied to infer fixed and random effects on the video behavior traits. The statistical model (1) was defined as follows:

$$y_{ijklmnopq} = \mu + sex_i + FG_j + BTS_k + SEAS_l + RBV - LS_m + b_1(W_n) + b_2(age_o) + pig_p + e_{ijklmnopq} \quad (1)$$

where $y_{ijklmnopqr}$ was the r^{th} video observation for REST, BCON, IFIGHT, FIGHT, RFIGHT, BITE, EXPLORE or RACT; μ was the overall mean; sex_i was the fixed effect of the i^{th} sex (male or female); FG_j was the fixed

Table 2
Number of pigs for the different sub-classes of backtest response and feeding group (CON= control group, HD= supplement of an herbal diet). (The frequencies within the feeding group x backtest score groups did not differ significantly with $P = 0.15$).

Backtest	Feeding group CON	HD	Total
Low resisters (LR)	16	24	40
Intermediate resisters (IR)	12	16	28
High resisters (HR)	22	14	36

effect of the j^{th} feeding group (HD or CON); BTS_k was the fixed effect of the k^{th} piglet backtest score (LR, IR or HR); $SEAS_l$ was the fixed effect for the l^{th} season (spring 2018, summer 2018, winter 2018, spring 2019) at recording; $RBV\text{-}LS_m$ was the fixed effect for the m^{th} RBV-LS class of the sire (below or above the average), W_n was weaning or post weaning weight as a covariate (linear regression); age_o was the age of the pig at the recording date as a covariate (linear regression); b_1 and b_2 were the regression coefficients of the trait on W_n and age_o, respectively; pig_p was the random permanent environmental effect of the p^{th} pig due to the repeated measurements; and $e_{ijklmnopq}$ was the random residual effect. As outlined in the data section, we additionally made video analyses 2 and 4 days after the weaning date. However, repeatability model applications with more than two repeated measurements per pig failed to converge properly. Because of the already very complex model, we tested the interactions of the main effects (feeding group, LS-group, backtest score) separately in consecutive runs. These interactions were not significant ($P > 0.05$), which was the reason for excluding the interactions from the final model 1.

From the same model 1 and using the same R function, least squares means (lsmeans) for all levels of fixed effects were computed. Pair wise comparisons were adjusted for multiple comparisons by applying the Tukey-Kramer test. Differences between levels within fixed effects were considered as significant at $P < 0.05$.

We were also interested in estimating effects of the sires representing nine Piétrain sire lines on video behavior traits. However, additional inclusion of the random sire effect into the statistical model (1) failed convergence due to the rather small phenotypic dataset. We attempted to circumvent this problem via a two-step method of analysis, and we used the solutions from model (1) to correct the phenotypic observations for fixed effects. Hence, the pre-corrected phenotype y* of a pig p standardized to a weight w of 16 kg at an age of 40 days was:

$$y_p* = y_p - - sex_i - FG_j - BTS_k - SEAS_l - RBV - LS_m$$

In a second step, we run a reduced sire model (2) just considering the random sire effect on the pre-corrected phenotype of pig y*:

$$y*_{st} = \mu + sire_s + e_{st} \tag{2}$$

with $y*_{st}$ denoting the pre-corrected phenotype, $sire_s$ denoting the random sire effect, and e_{st} denoting the random residual effect.

3. Results

3.1. Phenotypic distribution of video behavior pattern

Fig. 1A shows the proportion for video behavior pattern at day 1 after weaning for the different backtest scores, and Fig. 1B the respective phenotypic associations 5 weeks later. Phenotypically, there is a clear tendency for aggressive behavior in the early post-weaning phase including IFIGHT, FIGHT and BITE for pigs classified as highly resistant HR piglets. Also for EXPLORE, reflecting first contact behavior with other pen contemporaries, the duration (in relation to the total video recording time) was longer in HR than in LR or IR pigs. Consequently, pigs classified as IR or LR piglets spent more time with normal behavior reflecting calmness and without disturbing other pigs, i.e., as shown via higher duration percentages for REST. For feeding, drinking and excretion (as summarized in RACT), duration percentages were very similar for all three groups HR, IR and LR. The distribution of video

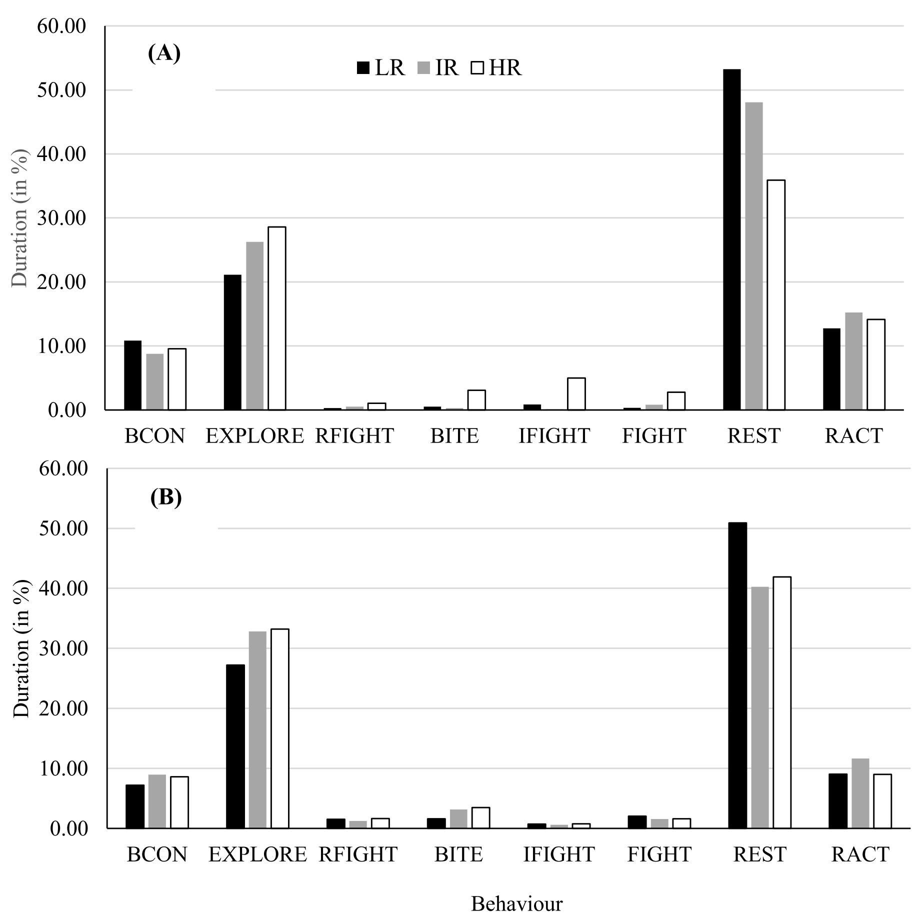

Fig. 1. Descriptive statistics for the duration of specific behaviors (in relation to the total time of 300 min) in the post-weaning period from video images within classes for piglet backtest scores (LR= low resisting piglets, IR= intermediate resisting piglets, HR= high resisting piglets) at the weaning date (A) and 5 weeks after weaning (B).

behavior scores displayed almost identical pattern when comparing both recording dates directly after weaning and 5 weeks later. However, the differences between the three backtest groups for the behaviors indicating aggressiveness (BITE, IFIGHT, FIGHT, EXPLORE) were negligible at the later video monitoring date after 5 weeks.

At both video recording dates at day 1 after weaning (Fig. 2A) and 5 weeks after weaning (Fig. 2B), pigs from the HD groups spent more time for REST than pigs from the CON group. Vice versa, for CON group pigs, we identified slightly longer periods for the aggressiveness indicators IFIGHT, FIGHT and BITE. Also first signs for body contact with other pigs (BCON) implied more time in the CON than in the HD groups. The HD supplement also had an effect on EXPLORE at the weaning date, contributing to a lesser extent of sniffing with the nose, touching the body of another pig with the snout, belly-nosing or genital nosing.

3.2. Significance of fixed effects and least squares means within fixed effect levels for video behavior pattern

The lsmeans for levels of classical fixed effects including sex and year-season of recording are given in Table 3. The duration for responses of most of the video behavior traits were very similar for male and female pigs and differences were non-significant ($P > 0.05$). Only for BCON, behavior for female pigs indicated stronger curiosity, with a significant ($P < 0.05$) longer duration of 31.60 min compared to 27.20 min for barrows.

Interestingly, during spring, pig behaviors indicating activity, aggressiveness and abnormality (FIGHT, BICON, BITE, EXPLORE), displayed significant longer durations compared to the remaining seasons. Consequently, the time spent for REST was smaller during spring. Regression coefficients for the age of the animal were negative for FIGTH, IFIGHT, BCON, BITE and EXPLORE, indicating a tendency for more calmness and longer duration of common behavior (eating, lying, sitting) in older pigs. At same ages, the heavier pigs were more aggressive, with longer duration for FIHGT, IFIGHT and BITE than younger pigs.

The lsmeans for video behavior traits within fixed effect levels for the backtest score and for the feeding group (Table 4) from the repeated measurement analyses (model 2) reflect the raw phenotypic means as depicted in Figs. 1 and 2. In this regard, the HR pigs spent significantly less minutes ($P < 0.05$) with REST than the LR pigs. Consequently, durations for all behavior pattern reflecting activity (IFIGHT, BCON, BITE, EXPLORE) were longer in HR than in LR pigs, with significant differences ($P < 0.05$) for the aggressiveness indicators FIGHT and BITE. The lsmeans for REST were larger in the HD than in the CON feeding group, indicating the significant effect of the herbal diet supplementation. The significantly longer duration for REST in the HD group was associated with less time spent for aggressive behavior including FIGHT, IFIGHT and BITE. Especially for FIGHT, the lsmeans in the HD group was extremely low with 0.08, and consequently, very sparely observed.

3.3. Effects of breeding values for lesion scores

With regard to RBV-LS of the sire (Table 4), we identified significant lsmeans differences ($P < 0.05$) for REST and for RFIGHT. The offspring of boars with RBV-LS above the threshold (indicating fewer skin lesions) spent more time for REST and RFIGHT compared to offspring of boars with RBV-LS below the threshold. Hence, the healthy genetic group (with regard to skin lesion) aimed on the activities reflecting calmness (lying, sitting, standing), and on avoiding fights. Same associations were identified when correlating the sire solutions (= estimated sire breeding values) from model (2) for the video behavior traits with their breeding values for lesion scores (Table 5). The Piétrain boars with favorable RBV-LS had breeding values below the average for video behavior traits indicating aggressiveness (FIGHT, BITE, IFIGHT) as indicated via negative breeding value correlations. In Table 4, we associated sire RBV-LS with phenotypic behavior pattern of offspring, and high RBV-LS were

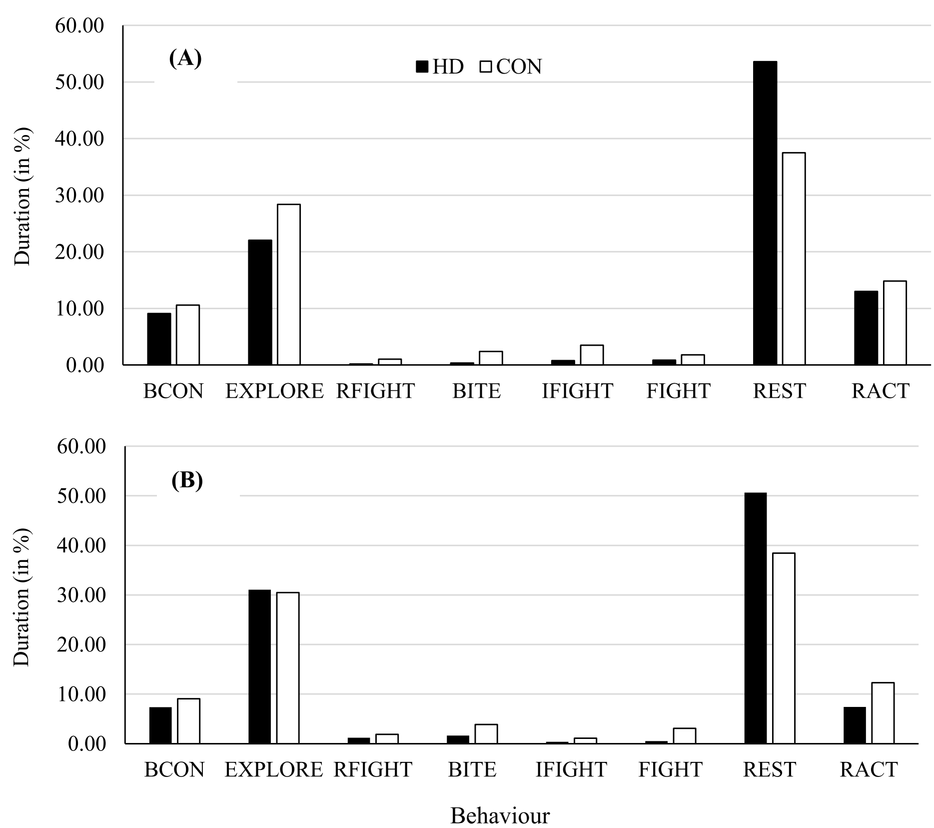

Fig. 2. Descriptive statistics for the duration of specific behaviors (in relation to the total time of 300 min) in the post-weaning period from video images within feeding classes (CON= control group; HD= supplementation of a herbal diet) at the weaning date (A) and 5 weeks after weaning (B).

Table 3
Least squares means (in minutes) within levels of fixed effects and regression coefficients for the covariate “age of animal” and “weaning weight” with corresponding standard errors (in parenthesis) for video behavior traits (significant differences at $P < 0.05$ for different levels within fixed effects are denoted with different superscripts).

Fixed effect	Behavior trait[1] REST	FIGHT	RFIGHT	IFIGHT	BCON	BITE	EXPLORE	RACT
Sex								
Male	140.00 (6.27)	4.59 (0.94)	1.11 (0.26)	6.02 (1.42)	27.20 (2.90)^a	3.06 (1.61)	76.30 (4.76)	43.70 (2.03)
Female	134.00 (6.79)	3.37 (1.02)	0.45 (0.31)	5.55 (1.54)	31.60 (3.15)^b	3.81 (1.75)	78.00 (5.17)	40.50 (2.21)
Season								
2018-spring	121.01 (12.37)^b	6.03 (1.85)^b	2.15 (1.02)	8.31 (2.80)	35.30 (5.71)^b	6.34 (3.17)^b	78.80 (9.39)	45.90 (4.01)
2018-summer	136.00 (11.73)^a	4.17 (1.77)^a	1.51 (0.97)	6.71 (2.67)	27.40 (5.45)^a	1.33 (3.02)^a	75.80 (8.95)	41.80 (3.82)
2019-winter	146.27 (7.94)^a	4.60 (1.20)^a	1.72 (0.65)	4.77 (1.81)	28.50 (3.69)^a	2.42 (2.05)^a	68.80 (6.06)	41.40 (2.59)
2019-spring	117.39 (9.28)^b	6.12 (1.39)^b	2.37 (0.76)	6.34 (2.11)	36.40 (4.30)^b	6.34 (2.39)^b	85.20 (7.06)	39.30 (3.01)
Age of animal	0.60 (2.49)	−0.36 (0.37)	0.24 (0.21)	−0.30 (0.56)	−0.43 (1.15)	−0.63 (0.64)	−1.41 (1.89)	0.90 (0.80)
Weaning weight	−3.50 (4.34)	0.85** (0.65)	−0.60 (0.36)	1.92 (0.98)	0.14 (2.01)	0.98 (1.11)	0 0.47 (3.29)	−1.65 (1.41)

[1] Abbreviations of behavior traits as defined in Table 1.

Table 4
Least squares means (in minutes) and corresponding standard errors (in parenthesis) for video behavior traits within levels of fixed effects (significant differences at $P < 0.05$ for different levels within fixed effects are denoted with different superscripts).

Fixed effect	Behavior trait[1] REST	FIGHT	RFIGHT	IFIGHT	BCON	BITE	EXPLORE	RACT
Backtest score[2]								
LR	157.06 (11.50)^a	3.39 (1.89)^a	4.96 (1.49)	1.64 (0.94)	22.72 (3.76)	5.82 (2.05)^a	81.59 (7.00)^a	26.91 (6.40)
IR	137.00 (17.50)^b	5.53 (2.54)^{a,b}	2.41 (2.27)	1.97 (1.41)	27.42 (5.72)	9.35 (3.12)^b	102.15 (7.64)^b	28.04 (8.96)
HR	136.62 (13.90)^b	8.34 (2.18)^b	2.75 (1.81)	2.88 (1.45)	26.31 (4.56)	10.73 (2.48)^b	104.28 (8.47)^b	25.81 (7.46)
RBV-LS-sire[3]								
High	156.00 (13.10)^a	5.08 (2.02)	1.17 (1.71)^a	1.96 (1.07)	26.63 (4.31)	5.74 (2.34)	90.21 (8.00)	26.55 (7.02)
Low	141.00 (10.60)^b	6.03 (1.96)	5.58 (1.38)^b	2.27 (0.85)	26.44 (3.47)	9.53 (1.89)	100.37 (6.46)	28.67 (6.61)
Feeding group[4]								
CON	130.00 (11.80)^a	11.02 (1.91)^a	3.78 (1.54)	3.20 (0.96)	27.61 (3.88)	10.89 (2.11)^a	90.7 (7.22)	31.82 (6.66)
HD	167.00 (11.30)^b	0.08 (1.94)^b	2.97 (1.47)	1.03 (0.09)	25.43 (3.70)	6.38 (2.01)^b	99.7 (6.87)	28.36 (6.41)

[1] Abbreviations of behavior traits as defined in Table 1.
[2] LR = low resisting piglets; IR = intermediate resisting piglets; HR = high resisting piglets.
[3] High = Relative breeding value for sire lesion score (LS) > 100; Low = relative breeding value for sire lesion score (LS) ≤ 100.
[4] CON = basal diet without supplements; HD = basal diet supplemented with a mixture of herbs.

Table 5
Correlations between relative breeding values for skin lesion scores (RBV-LS)[1] with breeding values for video behavior traits[2] and among breeding values for video behavior traits of Piétrain boars. Correlations significantly differing from zero (P < 0.05) are highlighted in bold.

Behavior trait	REST	FIGHT	RFIGHT	IFIGHT	BCON	BITE	EXPLORE	RACT
RBV-LS	0.41	−0.33	0.20	−0.14	−0.09	−0.36	−0.19	0.37
REST		−0.44	0.10	−0.28	−0.10	−0.20	−0.14	0.56
FIGHT			−0.18	0.51	0.22	0.28	0.27	−0.35
RFIGHT				−0.34	−0.11	−0.15	−0.05	0.38
IFIGHT					0.40	0.45	0.26	−0.39
BCON						0.23	0.21	−0.17
BITE							0.32	−0.47
EXPORE								−0.11

[1] Standardized relative breeding values above 100 are favorable, indicating less skin lesions than the average.
[2] Abbreviations of behavior traits as defined in Table 1. The breeding values are expressed in the unit of the traits (in minutes), indicating that larger breeding values are associated with longer duration for the respective trait.

associated with a short duration for RFIGHT. Oppositely, the sire breeding value correlations between RBV-LS and RFIGHT were positive with 0.20 (Table 5). Hence, the consideration of all related animals in the pedigree relationship might contribute to differences in phenotypic and genetic correlations, but opposite signs in this regard suggest more detailed investigations or analyses of additional data. Nevertheless, the breeding value correlation between RBV-LS and RFIGHT of 0.20 did not differ significantly from zero ($P > 0.05$).

Breeding value correlations between RBV-LS with breeding values for REST and RACT were positive, genetically indicating fewer skin lesions for pigs spending more time with “usual” and calm behavior (e.g., eating, drinking, lying, sitting and standing). The breeding value correlations among traits reflecting aggressiveness (FIGHT, IFIGHT, BITE) were positive, but these behaviors were negatively correlated with the traits indicating calmness (REST, RACT). The breeding value correlation between REST and RACT was positive with 0.56, being the largest correlation among all trait combinations. The breeding value for IFIGHT was negatively correlated with the breeding value for refusing fights (RFIGHT).

4. Discussion

4.1. The effect of feed supplements on behavior pattern in the post-weaning period

In the present study, the effects of the HD supplements on the video behavior pattern were very similar at day 1 after weaning and 5 weeks after weaning. Czycholl et al. (2019) studied the test-retest reliability of behaviors of growing pigs on-farm, which is the consistency of behavior pattern over time. Only for a few tests reflecting human-animal relationships and considering the first and second farm visit, the evaluation criteria indicated sufficient reliability. From a within-animal perspective, Botreau et al. (2013) indicated behavior consistencies with aging for uniform environmental characteristics, which are difficult to realize under field conditions. In our present study based on a standardized research design on-station, major herd environment characteristics were very similar at both behavior-monitoring dates, which might explain the almost identical behavior pattern within the HD and with the CON group for the two monitoring dates, even for the large interval between measurements of 5 weeks. The HD supplement seems to have a prompt effect at feeding day 2, indicating less fighting and biting, but longer durations for calm behavior signs as reflected via lsmeans for REST and RACT. Very prompt effects when supplementing the feeding ratio with herbs in terms of fewer skin lesions were reported by Aikins-Wilson et al. (2022), or more aggressive behavior for extremely strong feeding alterations, e.g., when switching from a liquid to a solid diet (Campbell et al., 2013). With regard to a solid feeding system, diets enriched with crude fiber contributed to less pig aggressiveness (Bernardino et al., 2016; Kallabis and Kaufmann, 2012), supporting the observations from the present study. Generally, an undersupply with specific feed ingredients seems to support abnormal behavior, as shown for increased ear and tail biting in a feeding group with reduced protein supply (Jensen et al., 1993; Fraser et al., 1991). With regard to the herbal supplement used in the present study, less aggressiveness in the HD compared to the CON group can be explained through the higher crude fiber (4.9% vs 2.4%) and crude protein content (17.1% vs 16.6%), plus the specific effects of the chicory herbs being a major component of the HD supplement. Pouille et al. (2022) reported favorable effects of chicory on different response variables (e.g., appetite regulation and health indicators) in a comprehensive mice study including metabolomics data and nutrigenomic analyses. In their review article, Street et al. (2013) addressed similar medical properties of the chicory ingredients from a plant perspective and for different species.

4.2. Associations between piglet behavior and behavior pattern in weaned pigs

The results from the present study indicate more aggressiveness after weaning (as indicated through long duration for FIGHT, BITE and IFIGHT) for piglets allocated to the HR group according to their backtest score. Hence, the active pigles with several escape attempts in the backtest also displayed activity at later ages including the abnormal activity pattern. Clouard et al. (2022) classified suckling piglets into 3 behavior categories reflecting 3 different social styles, i.e., low-solicited inactive animals (inactive), active animals (active) and highly-solicited avoiders (avoiders). With regard to repeated classifications over time, the group allocation of the individuals was quite stable, indicating that an active piglet at an early age was an active pig during aging. Furthermore, as identified in our present study, the effect of sex on behavior was not significant. In the study by Clouard et al. (2022), the group contemporaries did not change over time, but regrouping of piglets initiating a new social hierarchy within a group was identified as a major effect on agonistic behaviors (Fels et al., 2012). In the present study, aiming on a well-balanced research design with regard to several effects (genetic line, backtest score, RBV-LS), piglets were mixed after weaning. However, the barn environment in the research station did not change, i.e., the rearing environment and the post-weaning environment were identical with regard to farm location and farm employees. Hillmann et al. (2003) highlighted the environmental alterations from a feeding and husbandry perspective when evaluating behaviors in pigs. Studnitz et al. (2007) identified behavior changes in pigs with aging, e.g., a decrease of curiosity expressed by reduced sniffing and rooting, but changes in behavior pattern were similar for all individuals. Hence, a pig with longer duration for explorative behavior than herd contemporaries at a young age also displayed longer duration than the contemporaries at later ages, but on generally lower levels. In the present study, the lsmeans for EXPLORE and further traits reflecting pig activity were very similar at both recording dates in the post weaning period at day 1 after weaning and 5 weeks later. Consistent differences in behavior pattern from a between-animal perspective suggest the implementation of selection strategies at early ages.

The present study associated resistant piglet behavior with aggressiveness after weaning phenotypically. However, with regard to breeding and selection, it is imperative to estimate genetic correlations between early indicator and target traits, implying to broadening the study to a larger sample size. The small sample size in the present study indicates first associations, but validations based on larger datasets are needed. Nevertheless, the analysis of video images remains a challenge in this regard. In simulations, König and Swalve (2006) showed that early indirect selection is justified for genetic correlations with the breeding goal trait for genetic correlations larger than 0.70. Apart from the genetic correlations, some other factors including the costs and logistics of trait recording for the different traits at different ages, as well as generation intervals, should be taken into consideration when determining optimal selection strategies. From the logistic perspective, both trait recording schemes the piglet backtest as well as analyzing video images, are extremely labor and time intensive. As an alternative, sensor technique can be used to measure behavioral changes automatically. The sensor outputs were valuable in discriminating general behavior pattern in farm animals such as between eating, sleeping or estrus detection, but the differentiation within behavior categories was quite difficult (Jaeger et al., 2019). Enhanced camera technique considering locomotor activities from 3D trajectories are more accurate and allow a clearer behavior picture (Matthews et al., 2017), but to identify actors for tail biting implies the time consuming study of video images of a trained person. A promising approach to specify pig behavior in detail is the combination of video data combined with innovative statistical predictions such as machine learning or semi-supervised neural network analyses (Wuttke et al., 2020). In the neural network or machine learning algorithms to predict later behavior, also trait responses from suckling piglets could be included.

4.3. Associations between skin lesions and aggressive behavior: victim-actor perspective

Efficient prevention of an outbreak for aggressive behavior, especially for tail biting, implies the separation of biters from the group. First signs of tail lesions were associated with exponential increasing activities for tail biting (Statham et al., 2009). However, as stated above, identification of aggressive pigs implies time-consuming and difficult video analyses, or a combination of modern video technique with complex statistical predictions such as machine learning algorithms (D'Eath et al., 2014). Hunter et al. (2001) stated that the identification of victims is much easier than the detection of biters. Prunier et al. (2019) indicated the broad variety of biting characteristics, which complicates to harmonize biting recording schemes. Consequently, from a genetics perspective, Gentz et al. (2019) and Aikins-Wilson et al. (2022) suggested selection strategies based on tail or skin lesions, especially in the context of group selection. However, to our knowledge, genetic correlation estimates between biting (actor) and being a victim, are not known. From a genetic-statistical perspective, it is a challenge to estimate genetic correlations between traits with small additive genetic variances or heritabilities. The heritabiltities for active tail biting was only 0.05 (Breuer et al., 2005), and of being a victim only 0.06 (Canario and Flatres-Grall, 2018). In the present study, we correlated boar breeding values for a victim trait (RBV-LS) with breeding values for an actor trait (BITE), indicating less skin lesions when selecting against biting. However, the estimates based on a small dataset for the actor trait, and breeding value correlations can differ substantially from true genetic correlations in case of low breeding value accuracies (Calo et al., 1973). Nevertheless, results from some genomic studies confirm the breeding value correlation from the present study. Brunsberg et al. (2013) studied gene expressions in the brain of pigs, indicating very similar pattern for biters and for victims, but more differences in neutral pigs. Similarly, Wilson et al. (2012) identified same SNP markers with significant effects on biter and victim traits, but other SNPs were associated with being neutral. Hence, such molecular findings plus our estimate for the breeding value correlation indicate the possibility for indirect selection on victim traits to reduce biting.

Nevertheless, from a genetics-statistical perspective, it remains very difficult to infer genetic covariance components between biter and victim traits, due to mutual relationships between cause and effect. In this regard, the social component, i.e., the effect of biting on herd contemporaries, has to be taken into account, suggesting the application of genetic evaluations with social interaction effects (e.g., Ellen et al., 2014; Heidaritaba et al., 2019). In such context, Bijma et al. (2007) addressed the magnitude of covariances for residual effects between the individual of interest and the group members, which changed depending on the environmental characteristics. Canario and Flatres-Grall (2018) suggested direct selection against biting in case of direct-social correlations close to zero, due to negligible effects on the "to be bitten risk" of group members. Furthermore, for behavior traits in terms of "early learning behavior" or competition among siblings (Drake et al., 2008), it is imperative to consider the maternal-genetic component, but enhanced statistical modeling complicates to inferring genetic parameters and convergence of statistical models.

5. Conclusion

The multi-factorial semi-randomized experiment indicated significant effects of feeding (herbal diet), piglet behavior (backtest score) and sire breeding values for victim traits (RBV-LS) on behavior traits of pigs monitored via video images in the post-weaning period. The HD supplement implying a diet enriched with crude fiber, protein and chicory, was associated with less aggressive pig behavior, i.e. lower duration for FIGHT, IFIGHT and BITE compared to the CON group. The more active piglets indicated as highly resistant in the backtest showed more active and aggressive behavior pattern than piglets classified as LR or IR. Pigs with sire RBV-LS below the average indicating a higher lesion prevalence spent more time with explorative, fighting and biting behavior, but the differences between sire groups were not significant. The correlation between boar RBV-LS with breeding values for active biting was moderately negative. Consequently, selection according to favorable breeding values for skin lesions or early selection of IR or LR piglets might contribute to a reduction of aggressive behavior in crossbred fattening pigs. The comprehensive video analyses from two monitoring dates indicated a strong overlap of behavior signs (from a within-pig perspective) with aging. Genetic-statistical modelling approaches considering, e.g., interactions among main fixed effects simultaneously, which base on larger datasets, are strongly suggested for ongoing validations of the present findings.

Declaration of Competing Interest

We declare that we have no conflicts of interest

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